

On-Hardware AI Audit Report

SAMPLE — generated from canonical public-data benchmarks (PHM 2012 / NASA C-MAPSS / ETT). Customer fields illustrative. Every number traces to a versioned result file (Annex, p.7).

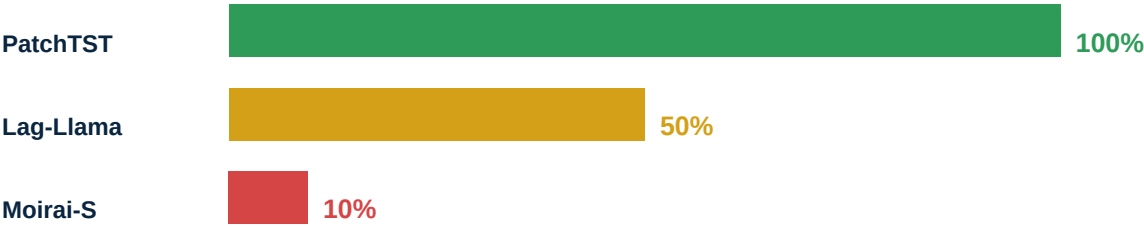
Customer	ACME Manufacturing (demo)	Audit window	4 weeks (protocol v0)
Hardware target	NVIDIA Jetson Nano 4GB	SLA budget	50 ms / inference
Task	Bearing PdM — RUL / anomaly	Action window	300 s (5 min)
Models in scope	8 foundation models (HF)	Data	PRONOSTIA PHM 2012 (public)

1 • Executive verdict

Same task, same data, same chip. What separates the models is not accuracy — it is **coverage**: the fraction of predictions that arrive in time, on the real hardware, under the real workload. Accuracy × coverage = events actually caught.

Model	Verdict	SLA coverage	Mean lead time	Rationale
PatchTST (ibm, 0.67M)	KEEP	1.00	260 s	Fastest inference; full coverage at cadence
Lag-Llama (2.45M)	OPTIMIZE	0.50	248 s	Half of acquisitions missed; quantize or relax cadence
Moirai-S (11.6M)	SWAP	0.10	190 s	9.5% coverage — misses ~90% of alarm windows
TimesFM-200M	EXCLUDED	—	—	Best MAE of catalog (0.006) but 2,039 ms — 40× over SLA

Coverage measured by missed-alarm simulation on 4 PRONOSTIA bearings ($\sigma=3$ alarm threshold, Jetson Nano profile, T4→Nano calibration ×10). Sources: results/missed_alarm/summary.json · results/benchmarks/2026-04-07/.



SLA coverage — fraction of acquisitions processed within budget · 10× gap between best and worst

2 · Scope & method

The audit follows the AlignEdge 5-phase protocol (playbook v0). It answers one question the standard MLOps stack does not: **does this model, on this chip, under this workload, meet the SLA — before go-live?**

Phase	What happens	Output
1 · Intake & scoping	Input spec validation, asset profile (AAS), scope letter	Signed scope
2 · On-hardware profiling	Latency distribution (100+ runs), RAM peak, 4-check quality gate	HW profile per model
3 · Drift injection	5 failure modes injected on customer data; detection latency measured	Drift signature
4 · Coverage & ranking	SLA coverage per model; TOPSIS multi-criteria vs candidate catalog	Verdicts
5 · Report & debrief	This document + debrief call + 30-day follow-up	Decision

Why this is not covered by existing tooling: offline eval measures accuracy on irrelevant hardware; shadow/canary discover latency failures in production, on real users; vendor tools (SageMaker Inference Recommender, Triton Model Analyzer) are single-vendor, one-off, and size instances or configs — they do not select models, measure coverage over a workload, or monitor drift.

3 · Asset profile (AAS)

Field	Value	Field	Value
Device	Jetson Nano 4GB (ARM A57 + 128-core Maxwell)	SLA	50 ms / inference
RAM budget	2,048 MB for model	Workload	1 acquisition / 10 s per bearing
Sensors	2× accelerometer 25.6 kHz	Action window	300 s before failure
Connectivity	Air-gapped (on-prem)	Safety context	Missed alarm ■ false alarm (20:1–100:1)

Asset profile encoded as AAS Type 1 JSON (IEC 63278) — machine-readable constraints feed the selection matrix directly.

4 • Hardware profiling results

Canonical benchmark: 8 foundation models × 2 datasets on NVIDIA T4 (total compute cost \$0.30). Jetson Nano latency estimated via ×10 calibration factor; the production audit measures on-device.

Model	Params	T4 mean	T4 p95	Nano est.	RAM	MAE (C-MAPSS)	vs SLA 50ms
PatchTST	0.67M	5.9 ms	6.2 ms	59 ms	14 MB	1.303	borderline
Lag-Llama	2.45M	7.8 ms	8.9 ms	78 ms	20 MB	n/a (zero-shot)	over
MOMENT-1-large	341M	29.7 ms	33.2 ms	297 ms	1,408 MB	0.114	over
Moirai-1.1-R-small	11.6M	53.7 ms	62.9 ms	537 ms	106 MB	n/a (zero-shot)	over
Moirai-1.1-R-large	305M	72.6 ms	73.9 ms	726 ms	2,360 MB	n/a	over + RAM
Chronos-t5-tiny	8.4M	166.8 ms	203.1 ms	1.7 s	45 MB	n/a	over
Chronos-t5-large	709M	830.0 ms	902.9 ms	8.3 s	2,771 MB	n/a	over + RAM
TimesFM-200M	200M	2,039 ms	2,110 ms	20 s	—	0.006 (best)	40× over

The headline finding: the most accurate model in the catalog (TimesFM, MAE 0.006) is 40× over the SLA. The least accurate viable model (PatchTST) is the only one that can hold the alarm cadence. **Accuracy rankings and deployability rankings are different rankings.** No leaderboard shows the second one — it depends on your chip and your SLA.

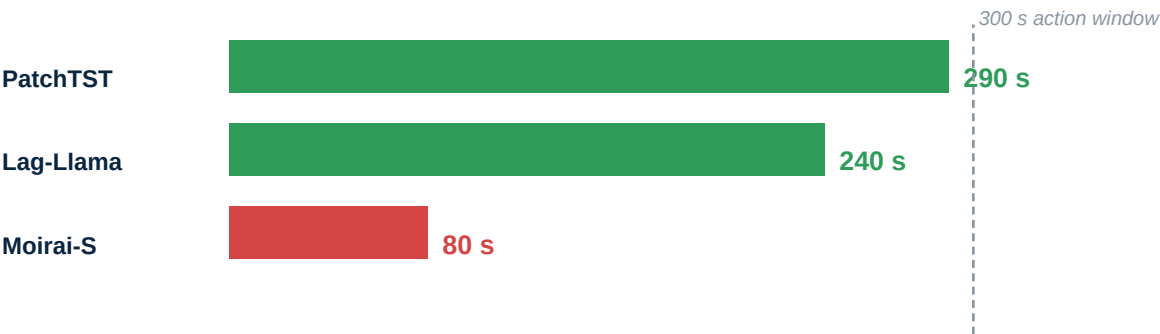
Latency-vs-SLA verdicts use per-inference latency; effective coverage at the 10 s acquisition cadence (p.4) is the operative metric — a model can exceed per-inference SLA yet keep partial cadence coverage by skipping acquisitions.

Quality gate: all 8 models passed shape/validity checks; zero-shot collapse (MAE=RMSE) screened; ONNX status recorded per model.

5 · Coverage analysis — the missed-alarm exposure

Coverage was measured by replaying 4 run-to-failure bearing traces (PRONOSTIA, PHM 2012 Challenge) through each model at the production cadence on the Jetson Nano profile. An alarm is **actionable** only if it fires with lead time inside the 300 s action window.

Model	Mean coverage	Mean lead time	Mean alarms fired	Spotlight: Bearing 3_2
PatchTST	1.00	260 s	39.2	85% coverage · 290 s lead
Lag-Llama	0.50	248 s	19.0	64% coverage · 240 s lead
Moirai-S	0.095	190 s	3.0	9.3% coverage · 80 s lead



LEAD TIME TO FAILURE — BEARING 3_2 · same task, same data · real bearing run-to-failure trace

Business translation. With a 99%-accurate model at 9.5% coverage, the fraction of failure events actually caught is 9.4%. Industry cost asymmetry for missed failures vs false alarms runs 20:1 to 100:1 (representative: \$25,000 vs \$500 per event). The coverage gap above is therefore not a performance nuance — it is the dominant term in the deployment's risk budget.

6 · Drift robustness — what happens after go-live

A model that passes the audit today degrades silently as signals drift. The audit characterizes each model's drift signature by injecting 5 canonical failure modes into the customer data and measuring detection behaviour of the DMCA-Drift v3.1.1 classifier (ADWIN + 8-feature decision tree).

Drift type	Detection accuracy (held-out)	Mean detection latency
Abrupt (step change)	100%	15 ticks
Variance shift	100%	38 ticks
Outlier-driven	100%	57 ticks
Incremental	60%	132 ticks
Gradual	100%	155 ticks
Overall (5 types × 5 seeds)	92% — false alarm rate 0.000	—

Closed-loop recovery (demonstrated on C-MAPSS): baseline MAE 0.796 → drift detected at MAE 3.121 → automatic re-selection (swap to Chronos-t5-tiny) → post-recovery MAE 0.174. Detection-to-recovery: 20 timesteps.

The false-alarm rate of 0.000 matters operationally: every false drift alarm triggers an unnecessary re-validation cycle. The classifier is calibrated to err on missed classification (reported honestly above: incremental drift at 60%) rather than false alarms.

This section is what converts a one-off audit into a runtime subscription: the certificate has an expiry date — continuous drift monitoring renews it.

7 • Recommendations

KEEP — PatchTST

Deploy as primary. Full coverage at cadence, 260 s mean lead. Accept MAE 1.303: at this SLA it is the ceiling of the accuracy-coverage trade-off. Enable drift monitoring (incremental drift is its weak spot).

OPTIMIZE — Lag-Llama

50% coverage is recoverable: (a) INT8 quantization (~2-3× latency reduction, to validate on-device), or (b) relax acquisition cadence to 20 s — halves required throughput at cost of 10 s extra detection delay. Re-audit after either change.

SWAP — Moirai-S

9.5% coverage misses ~90% of alarm windows; 0/4 bearings actionable. No optimization path closes a 10× gap. Recommended replacement: PatchTST (primary) or Chronos-t5-tiny as drift-recovery fallback (validated in closed-loop test).

EXCLUDED — TimesFM-200M

Keep out of the edge deployment; usable cloud-side for retrospective analysis where its MAE 0.006 is valuable and latency is irrelevant.

8 • Compliance annex (summary)

Every measurement in this report is logged with seed, timestamp and input hash (append-only JSONL) — aligned with EU AI Act Art. 12 logging and post-market monitoring duties for high-risk deployments. License inventory: all 8 models verified (Apache-2.0 / MIT); no copyleft exposure in the recommended configuration.

9 · Reproducibility annex

Artifact	Source file	Date
HW benchmark (8 models × 2 datasets, T4)	results/benchmarks/2026-04-07/benchmark_results_modal.csv	2026-04-07
Coverage / missed-alarm simulation	results/missed_alarm/summary.json · results.csv	2026-07-03
Drift held-out validation (v3.1.1)	results/drift_experiments/drift_holdout_2026-04-30_v3_1_1.json	2026-04-30
Closed-loop recovery test	results/drift_experiments/drift_cycle_results.json	2026-04-03
Audit protocol	docs/audit_playbook.md (v0)	2026-07-09

Datasets: PRONOSTIA (IEEE PHM 2012 Challenge, public) · NASA C-MAPSS FD001 (public) · ETT-h1 (public).
Seeds recorded per experiment. GPU compute cost of the full canonical benchmark: \$0.30 (NVIDIA T4, Modal).

About this sample. This report demonstrates the audit deliverable on public data. A production audit runs the same protocol on your model, your data sample, and your hardware node — and every number in sections 4-7 becomes yours. Protocol duration: 4 weeks.

AlignEdge — The right-sized AI model, on the right chip, at the right time.